



Statistics of Financial Markets

Chapter 7: Efficient markets, random walk and variance-ratio tests



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Acronyms used in this material (1/2)

- ▣ ACF: Autocorrelation Function
- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AMH: Adaptive Markets Hypothesis
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ CME: Chicago Mercantile Exchange
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ EMH: Efficient Market Hypothesis
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ EU: European Union
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GMM: Generalized Method of Moments
- ▣ KPSS: Kwiatkowski–Phillips–Schmidt–Shin (stationarity test)
- ▣ LB: Ljung–Box (test)
- ▣ MA: Moving Average (model)
- ▣ MIT: Massachusetts Institute of Technology
- ▣ OLS: Ordinary Least Squares
- ▣ PP: Phillips–Perron (unit-root test)

Acronyms used in this material (2/2)

- PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- RW1: Random Walk 1 (i.i.d. increments)
- RW2: Random Walk 2 (independent increments)
- RW3: Random Walk 3 (uncorrelated increments)
- SFE: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- VR: Variance Ratio
- WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's Question and Route

- **Question:** can yesterday's prices tell us anything about tomorrow's return?
 - ▶ if yes, a simple rule could beat the market; if no, the market is efficient in the weak sense
 - ▶ Chapter 4 defined the random walk and the martingale; here we test them on real prices
- **Route** of the chapter
 - ▶ efficient markets: three forms, the joint-hypothesis problem
 - ▶ random walk hypotheses RW1, RW2, RW3 and the martingale
 - ▶ white noise and the autocorrelation function; Ljung–Box and runs tests
 - ▶ unit roots: ADF, Phillips–Perron, KPSS; prices against returns
 - ▶ variance-ratio tests; efficiency across markets and over time; anomalies

Learning Outcomes

- ▣ State the three forms of market efficiency and explain why every test is a joint test
- ▣ Tell RW1, RW2, RW3 and the martingale apart, and say which of them volatility clustering violates
- ▣ Read a sample ACF with the right confidence band and compute Box–Pierce, Ljung–Box and runs statistics
- ▣ Run and combine ADF, Phillips–Perron and KPSS tests on prices and on returns
- ▣ Compute a variance ratio and its robust test, for one horizon and for several horizons together
- ▣ Judge, on real data, which markets look efficient, and whether this changes over time

Reading and Tools

- ▣ Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 11 (11.3 expectations and efficient markets, 11.5 random walk tests, 11.6 unit root tests)
 - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 11
- ▣ Econometrics of predictability: [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 2 (random walks, variance ratios) and Ch. 4 (event studies)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_07](#)
 - ▶ ported from the SFE Quantlets SFEtimewn, SFEacfar1, SFEacfar2, SFEacfma1 and SFEacfma2
- ▣ Lecture notebook: [open in Google Colab](#)
- ▣ Video course:  [Efficiency of cryptocurrency markets – A GMM-based analysis](#)

Can You Beat the Market?

- If past prices predicted future returns, a simple rule would earn money with no risk
 - ▶ traders would use the rule; their trades would move prices until the pattern disappears
- The evidence on professional investors
 - ▶ mutual funds, on average, did not beat a buy-and-hold portfolio after costs [Jensen \(1968\)](#)
 - ▶ few funds show skill large enough to cover their costs [Fama & French \(2010\)](#)
- A survey of the arguments for and against: [Malkiel \(2003\)](#)



The trading floor of the New York Stock Exchange, 1963

Photo: [Thomas J. O'Halloran \(1963\)](#), [Library of Congress](#);
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1900: Prices as a Random Walk

- Louis Bachelier, *Théorie de la spéculation* [Bachelier \(1900\)](#)
 - ▶ the first mathematical model of prices: the change of a price is a sum of many small random shocks
 - ▶ the expected gain of the speculator is zero
- [Kendall \(1953\)](#): 22 British price series behave like a “wandering” series
 - ▶ the changes from one week to the next look like independent draws
- [Roberts \(1959\)](#): simulated random walks show “patterns” that chartists see in real prices

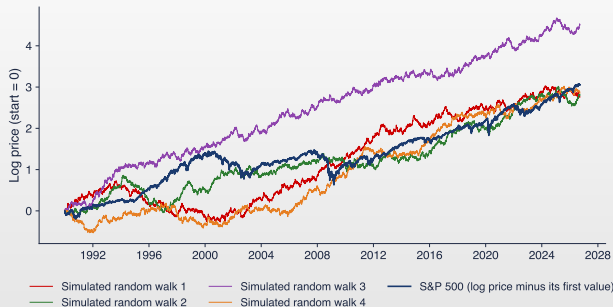


Louis Bachelier (1870–1946)

Photo: [unknown author](#); [public domain](#);

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Which One Is the S&P 500?



- Four random walks with i.i.d. Normal steps (mean 0.033%, standard deviation 1.13% per day, as the S&P 500) and the S&P 500 log price since 1990
- Question for the room:** without the legend, could you tell which path is real?
- Answer:** hardly; booms and long falls appear in pure noise too (end values: 2.76 to 4.50; S&P 500: 3.06)

1965: Properly Anticipated Prices Fluctuate Randomly

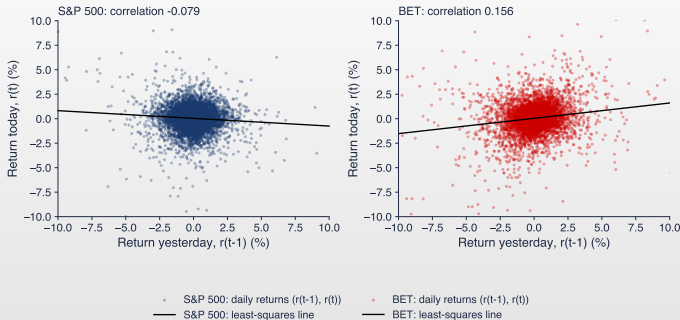
- Paul Samuelson [Samuelson \(1965\)](#): if prices already contain all that traders expect, the next change is unpredictable
 - ▶ formally: the price is a **martingale**, $E[P_{t+1} \mid \mathcal{F}_t] = P_t$ ([Chapter 4](#))
 - ▶ $\mathcal{F}_t$: the information available at time t
- Randomness of price changes is a sign of a market that works, not of chaos
- [Fama \(1965\)](#): daily returns of the 30 Dow Jones stocks show almost no autocorrelation



Paul A. Samuelson (1915–2009), Nobel Prize 1970

Photo: [Bernard Gotfryd \(1970–1975\)](#); public domain;
[Wikimedia Commons](#)

Does Yesterday Predict Today?



- Each point: (return yesterday, return today); the line: least squares; its slope is the first autocorrelation $\hat{\rho}(1)$
- S&P 500: $\hat{\rho}(1) = -0.079$ (a slight reversal); BET: $\hat{\rho}(1) = 0.156$ (a slight continuation); are these numbers different from zero?

Recap: Can Prices Be Predicted?

- From Bachelier (1900) to Samuelson (1965): price changes look random, and economics explains why
- Professional investors rarely beat the market after costs
- Daily autocorrelations are small but not zero: we need tests, not impressions

What Is an Efficient Market?

- **EMH** (efficient market hypothesis) **Fama (1970)**: prices “fully reflect” the available information Ω_t
 - ▶ no trading rule based on Ω_t earns more than the return that compensates its risk
 - ▶ prices move only on **news**: information that was not in Ω_t
- Efficiency is relative to an information set
 - ▶ the smaller Ω_t , the weaker the claim and the easier the test
- Efficient does not mean “returns are zero”: the expected return pays for risk



Eugene F. Fama, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia

[Commons](#)

Three Forms of Efficiency

Form	Information set Ω_t	What cannot earn excess returns	Typical test
Weak	past prices and returns	technical analysis, chart patterns	autocorrelation, runs, variance ratio
Semi-strong	all public information	fundamental analysis of public data	event studies
Strong	all information, also private	insider information	returns of insiders and fund managers

- The forms are nested: strong $\Rightarrow$ semi-strong $\Rightarrow$ weak; a rejection of the weak form rejects all three
- This chapter tests mainly the weak form: past prices are free data for everyone [Fama \(1970\)](#), [Fama \(1991\)](#)

The Fair-Game Model

- Let r_{t+1} be the return of an asset and $E[r_{t+1} | \Omega_t]$ its **equilibrium expected return** (the return that pays for its risk)
 - ▶ the **excess return** (surprise): $z_{t+1} = r_{t+1} - E[r_{t+1} | \Omega_t]$
- **Fair game**: $E[z_{t+1} | \Omega_t] = 0$
 - ▶ forecast errors are zero on average whatever we know at time t
 - ▶ a rule that invests $w(\Omega_t)$ earns $E[w(\Omega_t) z_{t+1}] = 0$ in excess of equilibrium
- With a constant expected return μ : $E[r_{t+1} | \Omega_t] = \mu$; returns minus μ are a **martingale difference**
- Nothing here says the variance is constant or the distribution is Normal

The Joint-Hypothesis Problem

- To test efficiency we must know the equilibrium expected return $E[r_{t+1} | \Omega_t]$
 - ▶ this needs a model of equilibrium: constant mean, CAPM (capital asset pricing model), factor models
- Every test of efficiency is a **joint test** of efficiency and of the model [Fama \(1991\)](#)
 - ▶ a rejection may mean an inefficient market, or a wrong model of expected returns
 - ▶ efficiency alone can never be rejected
- For daily returns the problem is small: daily expected returns are tiny (about 0.033% for the S&P 500), so any reasonable model gives almost the same answer

The Grossman–Stiglitz Paradox

- Information is costly to collect and to analyse [Grossman & Stiglitz \(1980\)](#)
 - ▶ if prices reflected all information perfectly, nobody would be paid for collecting it
 - ▶ then nobody would collect it, and prices could not reflect it
- Conclusion: a market can only be **nearly** efficient
 - ▶ small inefficiencies pay informed traders for their costs
 - ▶ the question becomes “how efficient, where and when?”, not “yes or no?”
- This view leads to the adaptive markets hypothesis of Section 8

2013: One Nobel Prize, Two Views



Eugene F. Fama: prices are hard to predict over days and weeks

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons



Robert J. Shiller: prices swing too much compared with fundamentals

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- Both are right, at different horizons: short-run returns are almost unpredictable; long-run returns are partly predictable from valuation ratios

Recap: Efficient Markets

- Efficiency is defined relative to an information set: weak, semi-strong, strong
- Fair game: excess returns are unpredictable given Ω_t
- Every test is a joint test with a model of expected returns
- Costly information makes perfect efficiency impossible Grossman & Stiglitz (1980)

The Random Walk Model of Log Prices

- Log price $p_t = \ln P_t$; **random walk with drift** μ : $p_t = \mu + p_{t-1} + \varepsilon_t$
 - ▶ the return $r_t = p_t - p_{t-1} = \mu + \varepsilon_t$: the increments ε_t have mean 0
 - ▶ by recursion: $p_t = p_0 + \mu t + \sum_{s=1}^t \varepsilon_s$: every shock is **permanent**
- If the ε_t are uncorrelated with variance σ^2 :
 - ▶ $E[p_t] = p_0 + \mu t$ and $\text{Var}(p_t - p_0) = t\sigma^2$: the uncertainty grows with the horizon
 - ▶ the q -day return $r_t(q) = r_t + \dots + r_{t-q+1}$ has $\text{Var}(r_t(q)) = q\sigma^2$
- Log prices, not prices: P_t cannot become negative, and log returns add over time ([Chapter 1](#))

Three Random Walk Hypotheses

- ▣ **RW1** (i.i.d. increments): ε_t independent and identically distributed, $\varepsilon_t \sim \text{IID}(0, \sigma^2)$ [Campbell, Lo & MacKinlay \(1997\)](#)
 - ▶ the strongest: no function of the past predicts any function of the future
- ▣ **RW2** (independent increments): ε_t independent, but not identically distributed
 - ▶ allows the variance to change over time in a way that is not predictable from the past (regimes)
- ▣ **RW3** (uncorrelated increments): $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ for all $k \neq 0$
 - ▶ the weakest: allows $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$, that is, volatility clustering
- ▣ Nested: $\text{RW1} \Rightarrow \text{RW2} \Rightarrow \text{RW3}$; tests of RW3 use only autocorrelations of returns

What Each Hypothesis Allows

Property of the returns	RW1	RW2	RW3	Martingale
$\text{Corr}(r_t, r_{t-k}) = 0$	yes	yes	yes	yes
$E[r_t \mid \text{past}] = \mu$ (unpredictable mean)	yes	yes	not always	yes
constant variance	yes	no	no	no
volatility clustering allowed	no	no	yes	yes
GARCH returns (Chapter 9) satisfy it	no	no	yes	yes

- **GARCH** (generalised autoregressive conditional heteroskedasticity): the variance of tomorrow depends on today's shocks; the mean stays unpredictable
- Weak-form efficiency needs only a martingale (or RW3), not RW1

Martingale and Random Walk

- $\{p_t - \mu t\}$ is a **martingale** if $E[p_{t+1} - \mu(t+1) \mid \mathcal{F}_t] = p_t - \mu t$ Samuelson (1965)
 - ▶ equivalently, $E[r_{t+1} - \mu \mid \mathcal{F}_t] = 0$: the returns minus μ are a martingale difference
 - ▶ a martingale difference with finite variance is uncorrelated: martingale $\Rightarrow$ RW3
- RW1 and RW2 imply the martingale; the converse fails: a martingale may have predictable variance
- **Question for the room:** does volatility clustering contradict weak-form efficiency?
 - ▶ **Answer:** no; it rejects RW1 and RW2, but a predictable variance gives no predictable return; the martingale and RW3 survive

Worked Example: How Risk Grows with the Horizon

- S&P 500 since 1990: daily standard deviation $\hat{\sigma} = 1.13\%$
 - ▶ under a random walk, $\text{sd}(r_t(q)) = \sigma\sqrt{q}$
- Step by step
 - ▶ one week, $q = 5$: $1.13 \times \sqrt{5} = 1.13 \times 2.24 = 2.53\%$
 - ▶ one month, $q = 20$: $1.13 \times 4.47 = 5.07\%$
 - ▶ one year, $q = 250$: $1.13 \times 15.81 = 17.92\%$
- This is the “square-root-of-time rule” used to scale VaR and volatility
- If returns are autocorrelated, the rule fails: the variance-ratio test of Section 7 measures by how much

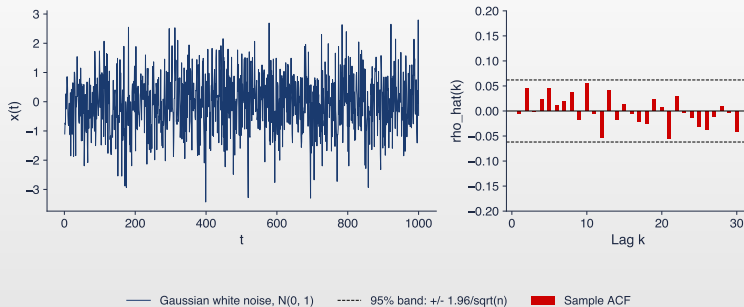
Recap: Random Walk Hypotheses

- ▣ $p_t = \mu + p_{t-1} + \varepsilon_t$: permanent shocks, $\text{Var}(r_t(q)) = q\sigma^2$
- ▣ RW1 (i.i.d.) $\Rightarrow$ RW2 (independent) $\Rightarrow$ RW3 (uncorrelated)
- ▣ Martingale: unpredictable mean, possibly predictable variance; the right null for efficiency

White Noise and Autocorrelation

- **Autocovariance** at lag k : $\gamma(k) = \text{Cov}(x_t, x_{t-k})$; **autocorrelation**: $\rho(k) = \gamma(k)/\gamma(0)$
 - ▶ the **ACF** (autocorrelation function): $k \mapsto \rho(k)$, $k = 1, 2, \dots$
- **White noise**: mean 0, constant variance σ^2 , $\rho(k) = 0$ for every $k \neq 0$
 - ▶ i.i.d. noise is white noise; Gaussian white noise is i.i.d. $N(0, \sigma^2)$
 - ▶ white noise need not be independent: GARCH returns are white noise but not i.i.d.
- **Sample ACF**: $\hat{\rho}(k) = \sum_{t=k+1}^T (x_t - \bar{x})(x_{t-k} - \bar{x}) / \sum_{t=1}^T (x_t - \bar{x})^2$
 - ▶ for i.i.d. data, $\hat{\rho}(k) \approx N(0, 1/T)$: 95% band $\pm 1.96/\sqrt{T}$

Gaussian White Noise and Its Sample ACF



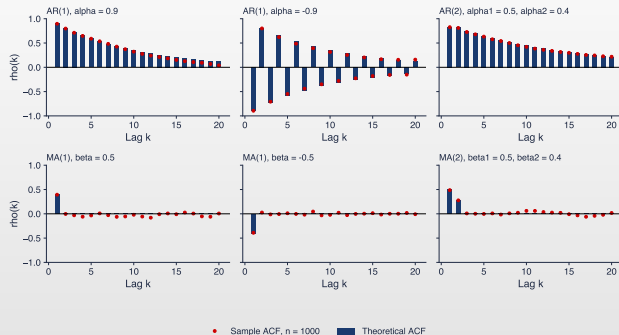
- Left: 1 000 draws from $N(0, 1)$ (as in SFEmtimewn); right: $\hat{\rho}(1), \dots, \hat{\rho}(30)$ with the band $\pm 1.96/\sqrt{1000} = \pm 0.062$
- Here 0 of 30 autocorrelations fall outside the band; with 30 lags we expect about 1.5 by chance alone

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The ACF of AR and MA Processes

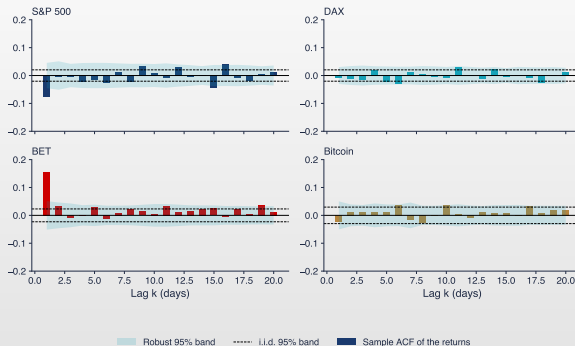
- ▣ **AR(1)** (autoregressive): $x_t = \alpha x_{t-1} + \varepsilon_t$, $|\alpha| < 1$: $\rho(k) = \alpha^k$
 - ▶ geometric decay; alternating signs if $\alpha < 0$; example: $\alpha = 0.9 \Rightarrow \rho(3) = 0.9^3 = 0.729$
- ▣ **AR(2)**: $x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \varepsilon_t$: $\rho(1) = \alpha_1 / (1 - \alpha_2)$, $\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2)$
 - ▶ $\alpha_1 = 0.5$, $\alpha_2 = 0.4$: $\rho(1) = 0.5/0.6 = 0.833$, $\rho(2) = 0.5 \times 0.833 + 0.4 = 0.817$
- ▣ **MA(1)** (moving average): $x_t = \varepsilon_t + \beta \varepsilon_{t-1}$: $\rho(1) = \beta / (1 + \beta^2)$, $\rho(k) = 0$ for $k \geq 2$
 - ▶ $\beta = 0.5$: $\rho(1) = 0.5/1.25 = 0.40$; an MA(q) has an ACF that cuts off after lag q
- ▣ The shape of the ACF tells the type of dependence: slow decay (AR), cut-off (MA), nothing (white noise)

Theoretical and Sample ACF of Six Processes



- Bars: theoretical $\rho(k)$; points: sample ACF of one simulated path of 1000 values (SFEacf1, SFEacf2, SFEacfma1, SFEacfma2)
- A random walk is the limit $\alpha \rightarrow 1$ of the AR(1): its ACF would not decay at all

The ACF of Daily Returns



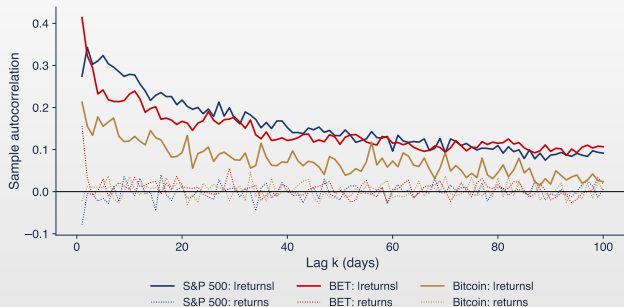
□ Bars: $\hat{\rho}(1), \dots, \hat{\rho}(20)$; dashed: the i.i.d. band $\pm 1.96/\sqrt{T}$; shaded: the robust band (next slide)

Reading the ACF of Returns

- First autocorrelations: S&P 500 -0.079 , DAX -0.009 , BET 0.156 , Bitcoin -0.022
 - ▶ only the S&P 500 and the BET stand out at lag 1, with opposite signs
- The **robust** band: $\pm 1.96 \sqrt{\hat{\delta}(k)/T}$, $\hat{\delta}(k) = T \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$, $e_t = r_t - \bar{r}$ Lo & MacKinlay (1988)
 - ▶ i.i.d. returns: $\hat{\delta}(k) \approx 1$; volatility clustering: large returns follow large returns, so $\hat{\delta}(k) > 1$
 - ▶ S&P 500 at lag 1: ± 0.046 instead of ± 0.020
- S&P 500, lags outside the band out of 20: 9 with the i.i.d. band, 3 with the robust band
- The i.i.d. band tests RW1; the robust band tests RW3, the hypothesis that matters for efficiency

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Returns Are Almost Uncorrelated, Their Size Is Not



- ACF of $|r_t|$: S&P 500 0.27 at lag 1, 0.22 at lag 20, 0.09 at lag 100; the returns themselves: dotted lines near 0
- Large moves follow large moves: RW1 and RW2 are rejected by eye; RW3 and the martingale are not touched by this chart

Recap: White Noise and the ACF

- White noise: uncorrelated; i.i.d. noise: independent; GARCH returns: white noise, not i.i.d.
- AR: geometric decay of the ACF; MA(q): cut-off after q lags
- Use the robust band $\pm 1.96\sqrt{\hat{\delta}(k)/T}$ for returns: the i.i.d. band is too narrow

Portmanteau Tests: Box–Pierce and Ljung–Box

- $H_0: \rho(1) = \dots = \rho(m) = 0$; H_1 : at least one $\rho(k) \neq 0$; one test for m lags together
 - ▶ “portmanteau”: a suitcase that carries many autocorrelations at once
- Box–Pierce [Box & Pierce \(1970\)](#): $Q_{BP}(m) = T \sum_{k=1}^m \hat{\rho}(k)^2$
- **LB** (Ljung–Box) [Ljung & Box \(1978\)](#): $Q_{LB}(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}(k)^2}{T-k}$, closer to χ^2 in small samples
 - ▶ under i.i.d. returns both $\sim \chi^2(m)$; reject H_0 if $Q > \chi_{0.95}^2(m)$; $\chi_{0.95}^2(10) = 18.31$
- Choice of m : here $m = 10$ (two trading weeks); a larger m dilutes a short-lived effect

Worked Example: Ljung–Box Step by Step

- ▣ $T = 1000$ daily returns, $\hat{\rho}(1) = 0.06$, $\hat{\rho}(2) = -0.04$, $\hat{\rho}(3) = 0.03$; test with $m = 3$
 - ▶ $T\hat{\rho}(k)^2$: 3.6, 1.6, 0.9; their sum: $Q_{BP}(3) = 6.10$
 - ▶ $Q_{LB}(3) = 1000 \times 1002 \times (0.06^2/999 + 0.04^2/998 + 0.03^2/997) = 6.12$
- ▣ Decision: $\chi_{0.95}^2(3) = 7.81 > 6.12$: do not reject H_0 at 5%
 - ▶ note: $\hat{\rho}(1) = 0.06$ lies just inside the band $\pm 1.96/\sqrt{1000} = \pm 0.062$; the joint test does not reject either
- ▣ The same numbers with $T = 5000$: Q five times larger, a clear rejection: small autocorrelations become significant in long samples

A Portmanteau Test Robust to Volatility Clustering

- The χ^2 law of Q_{LB} needs $\text{Var}(\hat{\rho}(k)) = 1/T$: true for i.i.d. returns, false with volatility clustering
 - ▶ then Q_{LB} rejects a true martingale far too often
- **Robust** statistic: divide each $\hat{\rho}(k)^2$ by its own variance $\hat{\delta}(k)/T$ Escanciano & Lobato (2009)
 - ▶ $\tilde{Q}(m) = \sum_{k=1}^m \frac{T\hat{\rho}(k)^2}{\hat{\delta}(k)} \sim \chi^2(m)$ for a martingale difference with finite fourth moment
 - ▶ with i.i.d. returns $\hat{\delta}(k) \approx 1$ and $\tilde{Q} \approx Q_{BP}$
- Rule: test RW1 with Q_{LB} ; test the martingale (weak-form efficiency) with $\tilde{Q}$

Autocorrelation Tests on Real Data

	T	$\hat{\rho}(1)$	$Q_{LB}(10)$	p	$\tilde{Q}(10)$	p	$Q_{LB}(10), r^2$	runs z	p
S&P 500	9 240	-0.079	90.8	< 0.001	18.8	0.043	8015	2.97	0.003
DAX	9 288	-0.009	21.8	0.016	8.6	0.571	3747	1.73	0.084
BET	7 243	0.156	198.3	< 0.001	42.9	< 0.001	2500	-7.75	< 0.001
Bitcoin	4 384	-0.022	20.2	0.028	11.9	0.294	213	2.68	0.007
TLV	3 788	0.010	29.1	0.001	13.2	0.210	481	-1.68	0.093
SNP	3 815	-0.015	20.7	0.023	9.9	0.446	505	1.72	0.085
BRD	3 771	0.032	19.9	0.030	10.8	0.375	416	-3.04	0.002
TGN	3 725	0.055	28.7	0.001	12.3	0.263	639	1.49	0.136

- Classic LB: all eight series reject at 5%; robust $\tilde{Q}$: only the S&P 500 ($p = 0.043$) and the BET ($p = < 0.001$)
- Squared returns: huge Q_{LB} everywhere: RW1 and RW2 are rejected for every series (volatility clustering)
- Start of the series: S&P 500 and DAX 1990; BET 1997; Bitcoin 2014; BVB stocks 2010 (TLV without 30–31 May 2016, a data error, [Chapter 2](#))

The Runs Test

- ▣ A **run**: a sequence of returns with the same sign; $++-++++--$ has 4 runs Wald & Wolfowitz (1940)
 - ▶ the test uses only the signs: it is **nonparametric**, insensitive to heavy tails and to volatility
- ▣ With n_1 positive and n_2 negative returns, $n = n_1 + n_2$, under independence:
 - ▶ $E[R] = \frac{2n_1n_2}{n} + 1$, $\text{Var}(R) = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}$, $z = \frac{R - E[R]}{\sqrt{\text{Var}(R)}} \approx N(0, 1)$
- ▣ Too few runs ($z < 0$): signs persist, positive dependence; too many runs ($z > 0$): signs alternate
 - ▶ BET: 3 278 runs against 3 606 expected, $z = -7.75$; S&P 500: 4 738 against 4 596, $z = 2.97$
 - ▶ the same directions as $\hat{\rho}(1)$: continuation in the BET, reversal in the S&P 500

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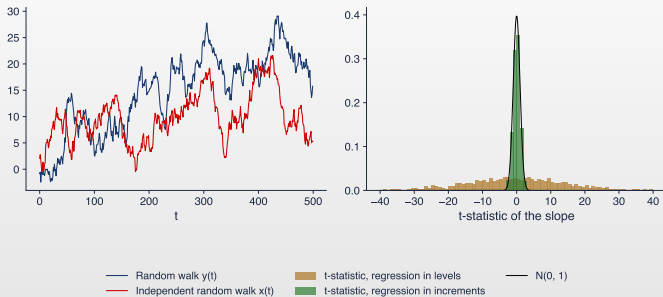
Recap: Testing for Autocorrelation

- ▣ $Q_{LB}(m) = T(T+2) \sum \hat{\rho}(k)^2 / (T-k) \sim \chi^2(m)$ under i.i.d. returns
- ▣ With volatility clustering use the robust $\tilde{Q}$: the verdict changes for six of eight series
- ▣ Runs test: signs only; too few runs means continuation
- ▣ $\tilde{Q}$: reversal in the S&P 500, continuation in the BET; no rejection for the DAX, Bitcoin and the BVB stocks

Stationary and Integrated Series

- **(Weakly) stationary** series: constant mean, constant variance, $\text{Cov}(x_t, x_{t-k})$ depends only on k
 - ▶ shocks fade away; the series returns to its mean; notation $I(0)$
- **Integrated of order 1**, $I(1)$: not stationary, but its first difference $\Delta x_t = x_t - x_{t-1}$ is
 - ▶ the random walk is $I(1)$: p_t wanders, $\Delta p_t = r_t$ is stationary
- **Unit root**: in $p_t = \phi p_{t-1} + \varepsilon_t$ the case $\phi = 1$; $|\phi| < 1$ gives a stationary AR(1)
 - ▶ the name: the root of $1 - \phi z = 0$ is $z = 1/\phi = 1$
- Question of this section: are log prices $I(1)$ and returns $I(0)$?

Why It Matters: Spurious Regression



- Regress one random walk on another, independent one ($T = 500$, 2 000 repetitions) [Granger & Newbold \(1974\)](#): $|t| > 1.96$ in 90% of cases, median $R^2 = 0.17$
- On the increments: $|t| > 1.96$ in 5.5% of cases, as it should be; lesson: test for unit roots before regressing levels ($|t| > 40$ in 2.6% of cases, not drawn)

The Dickey–Fuller Test

- Subtract p_{t-1} : $\Delta p_t = c + \gamma p_{t-1} + \varepsilon_t$ with $\gamma = \phi - 1$ [Dickey & Fuller \(1979\)](#)
 - ▶ H_0 : $\gamma = 0$ (unit root); H_1 : $\gamma < 0$ (stationary); a one-sided test
 - ▶ statistic: $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$, the usual t -ratio of OLS (ordinary least squares)
- Under H_0 , τ does **not** follow a t or Normal law: its critical values are more negative
 - ▶ 5% critical values [MacKinnon \(1996\)](#): -2.86 with a constant, -3.41 with a constant and a trend (Normal: -1.645)
- Three specifications: no constant; constant c ; constant and trend $c + bt$
 - ▶ log prices drift upwards: use constant and trend; returns: constant only

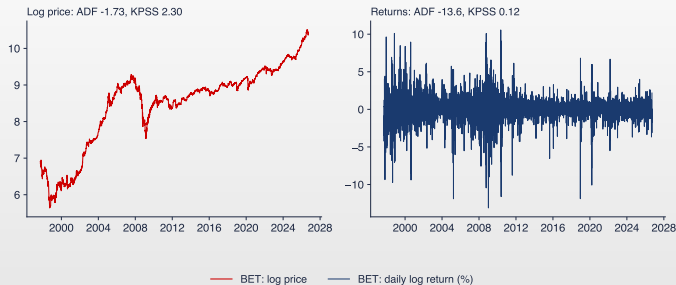
The Augmented Test and a Worked Example

- ▣ **ADF** (augmented Dickey–Fuller) [Said & Dickey \(1984\)](#): add lagged differences to remove autocorrelation from ε_t
 - ▶ $\Delta p_t = c + bt + \gamma p_{t-1} + \sum_{j=1}^L \delta_j \Delta p_{t-j} + \varepsilon_t$; same τ , same critical values
 - ▶ L chosen by AIC (Akaike information criterion, [Chapter 6](#)), at most $12(T/100)^{1/4}$
- ▣ **Worked example**: BET log price, constant, no lags, $T = 7\,243$
 - ▶ OLS: $\hat{\gamma} = -0.00001$, $\text{SE}(\hat{\gamma}) = 0.00016$
 - ▶ $\tau = -0.00001/0.00016 = -0.09 > -2.86$: do not reject the unit root
 - ▶ a usual t -test (critical value -1.645) would have rejected it: the wrong distribution gives the wrong answer

Phillips–Perron and KPSS

- **PP** (Phillips–Perron) [Phillips & Perron \(1988\)](#): the Dickey–Fuller regression without lags; τ is corrected for autocorrelation and heteroskedasticity
 - ▶ the correction uses a long-run variance with Bartlett weights [Newey & West \(1987\)](#); same H_0 and critical values as ADF
- **KPSS** (Kwiatkowski–Phillips–Schmidt–Shin) [Kwiatkowski et al. \(1992\)](#): the hypotheses are **reversed**
 - ▶ H_0 : stationary (around a constant or a trend); H_1 : unit root
 - ▶ $KPSS = \sum_{t=1}^T S_t^2 / (T^2 \hat{\lambda}^2)$, S_t the partial sums of the residuals, $\hat{\lambda}^2$ their long-run variance
 - ▶ reject stationarity if $KPSS > 0.463$ (constant) or > 0.146 (trend), at 5%
- Use both: ADF/PP ask “is there evidence against a unit root?”; KPSS asks “is there evidence against stationarity?”
 - ▶ ADF does not reject and KPSS rejects: $I(1)$; ADF rejects and KPSS does not: $I(0)$; both or neither reject: inconclusive

The BET: Log Price and Returns



- Log price: ADF -1.73 ($p = 0.736$), KPSS 2.30 ($p < 0.01$): a unit root, not stationary
- Returns: ADF -13.65 ($p < 0.001$), KPSS 0.12 ($p > 0.10$): stationary
- The 2008 fall looks like a temporary deviation, but the tests say: permanent shocks, no return to a trend line

Unit-Root Tests on Prices and Returns

	log price (constant, trend)			returns (constant)		
	ADF	PP	KPSS	ADF	PP	KPSS
S&P 500	-1.47	-1.57	2.23	-17.61	-105.05	0.12
DAX	-2.69	-2.65	0.94	-23.30	-97.40	0.04
BET	-1.73	-1.54	2.30	-13.65	-74.47	0.12
Bitcoin	-1.48	-1.54	1.81	-20.16	-67.84	0.13
TLV	-3.46	-3.35	0.57	-12.96	-60.79	0.23
SNP	-0.80	-0.74	2.50	-19.50	-62.71	0.29
BRD	-2.18	-2.19	1.64	-59.46	-59.53	0.29
TGN	-0.96	-0.92	0.71	-39.90	-57.83	0.22

- 5% critical values: ADF and PP -3.41 (trend), -2.86 (constant); KPSS 0.146 (trend), 0.463 (constant)
- Log prices: no ADF/PP rejection except TLV (ADF $p = 0.044$); KPSS rejects stationarity for all: prices are $I(1)$
- Returns: ADF and PP reject strongly; KPSS never rejects ($p > 0.10$): returns are $I(0)$

A Unit Root Is Not Proof of Efficiency

- ▣ A unit root in prices is **necessary** for a random walk, but **not sufficient**
 - ▶ example: $r_t = 0.3 r_{t-1} + \varepsilon_t$ makes p_t an $I(1)$ series whose increments are predictable
 - ▶ ADF cannot distinguish it from a random walk; the BET has a unit root and autocorrelated returns
- ▣ A stationary price (no unit root) would contradict efficiency: prices would return to a predictable level
 - ▶ TLV: ADF rejects at 5% with a trend, KPSS also rejects: an inconclusive case, not a profitable rule
- ▣ Unit-root tests have low power against ϕ close to 1: the efficiency question is about the increments, and needs tests on returns

Recap: Unit Roots

- ▣ $I(1)$: shocks are permanent; $I(0)$: shocks fade; random walk = $I(1)$ prices, $I(0)$ returns
- ▣ ADF and PP: H_0 unit root, Dickey–Fuller critical values; KPSS: H_0 stationarity
- ▣ Our data: log prices $I(1)$, returns $I(0)$; regressions in levels can be spurious
- ▣ A unit root is necessary, not sufficient, for a random walk

The Idea of the Variance Ratio

- Under a random walk, $\text{Var}(r_t(q)) = q \text{Var}(r_t)$; the **VR** (variance ratio) compares the two sides
 - ▶ $\text{VR}(q) = \frac{\text{Var}(r_t(q))}{q \text{Var}(r_t)} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho(k)$ Lo & MacKinlay (1988), Campbell, Lo & MacKinlay (1997)
- Reading the number
 - ▶ $\text{VR}(q) = 1$: random walk (RW1, RW2, RW3 all give 1)
 - ▶ $\text{VR}(q) > 1$: positive autocorrelation, **momentum** (moves continue)
 - ▶ $\text{VR}(q) < 1$: negative autocorrelation, **mean reversion** (moves are partly undone)
- One number summarises $q - 1$ autocorrelations with declining weights: more power against persistent, small autocorrelations

Worked Example: VR from the ACF of the S&P 500

- $\hat{\rho}(1), \dots, \hat{\rho}(4)$ of the S&P 500: $-0.079, -0.004, -0.006, -0.022$
 - ▶ $q = 2$: $\text{VR}(2) = 1 + \hat{\rho}(1) = 0.921$
 - ▶ $q = 5$: weights $2(1 - k/5) = 1.6, 1.2, 0.8, 0.4$: $\text{VR}(5) = 1 + 1.6\hat{\rho}(1) + 1.2\hat{\rho}(2) + 0.8\hat{\rho}(3) + 0.4\hat{\rho}(4) = 0.855$
- The Lo–MacKinlay estimator on the same data: $\text{VR}(2) = 0.922, \text{VR}(5) = 0.856$
 - ▶ almost the same numbers: the two forms differ only by small-sample corrections
- Meaning: the weekly variance is about 0.856 of five daily variances: daily moves of the S&P 500 are partly reversed within a week

Lo and MacKinlay (1988): Estimator and Tests

- With T returns, mean $\hat{\mu}$ and overlapping q -day sums Lo & MacKinlay (1988):

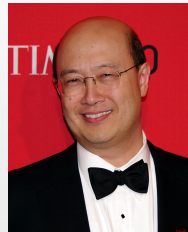
- $\hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t (r_t - \hat{\mu})^2$, $\hat{\sigma}_c^2(q) = \frac{1}{m} \sum_t (r_t(q) - q\hat{\mu})^2$

- $m = q(T - q + 1)(1 - q/T)$ corrects the bias; $\widehat{VR}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$

- Homoskedastic test (RW1):** $Z(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0, 1)$

- Robust test (RW3, martingale):** $Z^*(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{\hat{\theta}(q)/T}} \approx N(0, 1)$

- $\hat{\theta}(q) = \sum_{j=1}^{q-1} [2(q-j)/q]^2 \hat{\delta}(j)$, with the $\hat{\delta}(j)$ of the robust ACF band

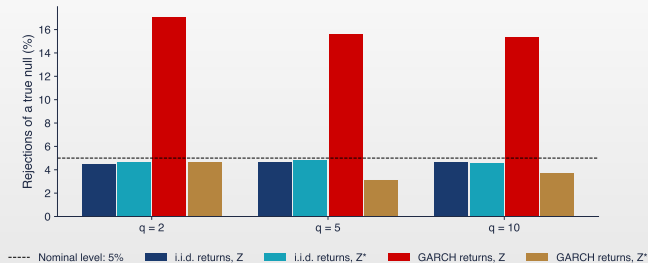


Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0;

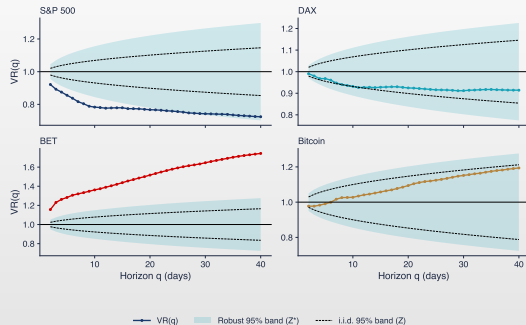
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 SFM_ch7_variance_ratio

Why the Robust Test: a Monte Carlo Check



- 1 000 samples of $T = 2000$ returns with H_0 true: i.i.d. Normal, and GARCH(1,1) ($\alpha = 0.12$, $\beta = 0.86$): uncorrelated, with volatility clustering
- i.i.d.: both reject about 5%; GARCH: $Z(2)$ rejects 17.1% of the time, $Z^*(2)$ only 4.7%
- The size of a test: how often it rejects a true H_0 ; Z finds “inefficiency” that is only volatility clustering

Variance Ratios for Horizons of 2 to 40 Days



- S&P 500: VR falls below 1 at every horizon (mean reversion); BET: VR rises up to about 1.517 at 20 days and beyond (momentum)
- DAX and Bitcoin: inside the robust band (shaded) at every horizon; the i.i.d. band (dashed) is much narrower

Variance Ratios and Robust Tests

	VR(2)	VR(5)	VR(10)	VR(20)	$Z^*(2)$	$Z^*(5)$	$Z^*(10)$	$Z^*(20)$
S&P 500	0.922	0.856	0.783	0.767	-3.37	-2.76	-2.73	-2.06
DAX	0.990	0.968	0.932	0.925	-0.61	-0.88	-1.21	-0.91
BET	1.156	1.283	1.362	1.517	5.99	5.25	4.74	4.94
Bitcoin	0.977	0.991	1.027	1.094	-0.88	-0.19	0.37	0.91
TLV	1.009	0.997	0.954	0.995	0.34	-0.04	-0.48	-0.04
SNP	0.985	0.960	0.975	1.038	-0.53	-0.68	-0.29	0.32
BRD	1.032	1.083	1.078	1.107	1.28	1.55	0.97	0.96
TGN	1.055	1.150	1.142	1.166	2.05	2.60	1.69	1.44

- $|Z^*| > 1.96$: reject the random walk at 5% for that horizon
- S&P 500: significant mean reversion at all four horizons; BET: significant momentum at all four
- Transgaz: $Z^*(2) = 2.05$ and $Z^*(5) = 2.60$; DAX, Bitcoin, TLV, SNP, BRD: no rejection
- But eight tests per series at once: some “significant” horizons appear by chance

Several Horizons at Once: the Chow–Denning Test

- ▣ Testing $q = 2, 5, 10, 20$ separately at 5% gives a much higher chance of at least one false rejection
 - ▶ for 4 independent tests: $1 - 0.95^4 = 18.5\%$
- ▣ **CD** (Chow–Denning) statistic [Chow & Denning \(1993\)](#): $CD = \max_i |Z^*(q_i)|$, $i = 1, \dots, m$
 - ▶ critical value from the studentised maximum modulus law: $P(CD \leq c) = (2\Phi(c) - 1)^m$
 - ▶ $m = 4$, 5%: $c = 2.49$ instead of 1.96; Φ : the standard Normal distribution function
- ▣ Reject the random walk if the largest $|Z^*|$ exceeds 2.49: one decision for all horizons

An Automatic Choice of the Horizon

- The horizon q can be chosen from the data: the automatic VR test [Choi \(1999\)](#)
 - ▶ $VR(k) = 1 + 2 \sum_{i=1}^{T-1} w(i/k) \hat{\rho}(i)$, with smoothly declining weights w (quadratic spectral kernel) [Andrews \(1991\)](#)
 - ▶ the bandwidth k grows with the first-order autocorrelation and with T ; statistic $\sqrt{T/k} (VR(k) - 1) / \sqrt{2} \approx N(0, 1)$
- **Wild bootstrap** p-value [Kim \(2009\)](#): resample $r_t^* = \eta_t(r_t - \bar{r})$ with $\eta_t \sim N(0, 1)$, 500 times
 - ▶ each r_t^* keeps the size of r_t (its volatility) but loses any autocorrelation: a robust null distribution
- Three tools, one question: $Z^*(q)$ for a chosen horizon, CD for a set of horizons, the automatic test for a horizon chosen by the data

Joint and Automatic Tests on Real Data

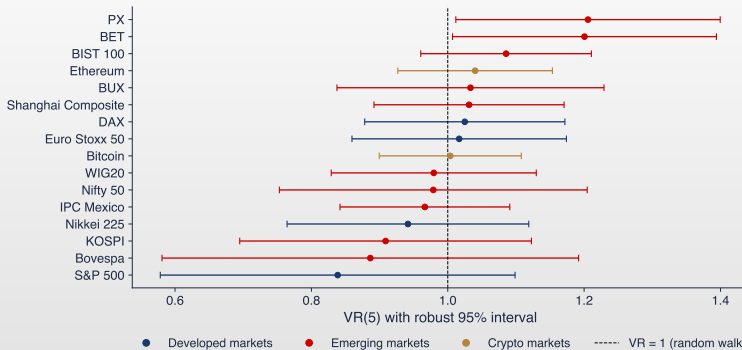
	T	CD	p	$\hat{k}$	automatic VR	p (bootstrap)
S&P 500	9 240	3.37	0.003	3.7	−5.21	0.006
DAX	9 288	1.21	0.644	1.7	−0.39	0.608
BET	7 243	5.99	< 0.001	5.6	8.86	< 0.001
Bitcoin	4 384	0.91	0.835	2.0	−1.01	0.352
TLV	3 788	0.48	0.981	1.5	0.31	0.718
SNP	3 815	0.68	0.937	1.7	−0.53	0.562
BRD	3 771	1.55	0.403	2.3	1.58	0.152
TGN	3 725	2.60	0.037	3.0	3.78	0.004

- Both tests reject for the S&P 500, the BET and Transgaz; neither rejects for DAX, Bitcoin, TLV, SNP and BRD
- $\hat{k}$ is large where the first autocorrelation is large (BET 5.6), small where it is near 0 (DAX 1.7)
- Interpretation: rejection does not mean profit: an autocorrelation of −0.079 predicts less than 1% of the variance of tomorrow's return; trading costs absorb most of it

Recap: The Variance-Ratio Test

- ▣ $VR(q) = 1 + 2 \sum_{k < q} (1 - k/q) \rho(k)$; > 1 momentum, < 1 mean reversion
- ▣ Use $Z^*(q)$: the homoskedastic $Z(q)$ over-rejects under volatility clustering
- ▣ Several horizons: Chow–Denning; data-driven horizon: automatic VR with wild bootstrap
- ▣ S&P 500 mean-reverting, BET trending; DAX, Bitcoin and most BVB stocks consistent with a random walk

Sixteen Markets over the Last Ten Years



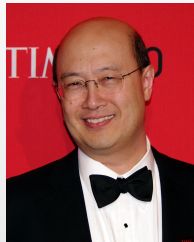
- VR(5) with its robust 95% interval, from 19 September 2016 to 18 September 2026: developed, emerging and crypto markets

Reading the Cross-Market Comparison

- ▣ Only two intervals exclude 1: the PX ($Z^*(5) = 2.08$) and the BET ($Z^*(5) = 2.03$), both above 1
 - ▶ two smaller Central European markets, with less liquid index stocks
- ▣ **Many markets, many tests:** 16 tests at 5% give 0.8 false rejections on average
 - ▶ if the tests were independent, the chance of at least one false rejection would be $1 - 0.95^{16} = 56\%$
 - ▶ Chow–Denning over four horizons: no market rejects at 5% (BET $p = 0.159$, PX $p = 0.142$)
- ▣ S&P 500: $VR(5) = 0.84$, the lowest, but with a wide interval ($Z^* = -1.22$): the large swings of 2020 inflate $\hat{\theta}$
- ▣ Bitcoin and Ethereum: $VR(5)$ close to 1, as for developed markets

The Adaptive Markets Hypothesis

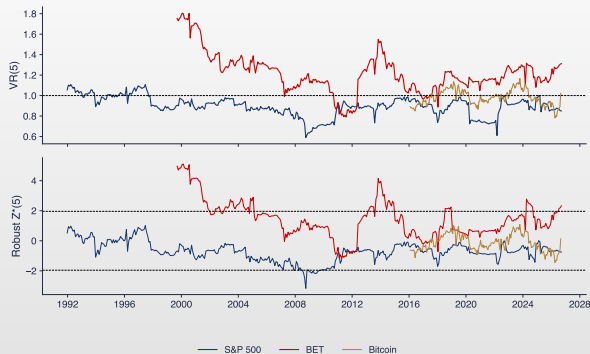
- **AMH** (adaptive markets hypothesis) [Lo \(2004\)](#): markets are ecosystems of traders who learn and adapt
 - ▶ profit opportunities appear, are exploited and disappear; efficiency changes with the environment
 - ▶ crises, new technology, new participants and regulation move the market away from efficiency or back towards it
- Testable consequence: predictability varies over time
 - ▶ evidence: [Kim, Shamsuddin & Lim \(2011\)](#)(U.S., a century of data); [Urquhart & McGroarty \(2016\)](#); survey: [Lim & Brooks \(2011\)](#)
- Tool: the same test on rolling windows



Andrew W. Lo, author of the AMH

Photo: [David Shankbone \(2012\)](#); CC BY 3.0;
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Rolling Variance Ratios



- $VR(5)$ and $Z^*(5)$ on windows of 500 trading days (about two years), a new window every 21 days, dated at the window end

Reading the Rolling Variance Ratios

- ▣ BET: VR(5) up to 1.80 in the window ending in 2000; closer to 1 after 2008, apart from short episodes
 - ▶ 29% of the windows reject at 5%, all of them with $VR > 1$ (momentum)
- ▣ S&P 500: VR(5) mostly below 1; the lowest, 0.59, in the window ending in 2008
 - ▶ 4% of the windows reject, all with $VR < 1$: reversals are strongest in crises
- ▣ Bitcoin: no window rejects since 2016 (185 windows)
- ▣ Caution: consecutive windows share 479 of 500 days; the share of rejecting windows is a description, not a test

Three Sub-Periods per Market

	period	T	$\hat{\rho}(1)$	VR(5)	$Z^*(5)$	CD p
BET	1997–2006	2 306	0.270	1.50	6.15	< 0.001
BET	2007–2016	2 516	0.066	1.08	0.82	0.262
BET	2017–2026	2 421	0.064	1.20	2.04	0.154
S&P 500	1990–2001	3 025	0.017	0.96	−0.62	0.334
S&P 500	2002–2013	3 019	−0.100	0.80	−2.53	0.010
S&P 500	2014–2026	3 196	−0.124	0.85	−1.30	0.080
Bitcoin	2014–2017	1 201	0.028	0.98	−0.23	0.983
Bitcoin	2018–2021	1 461	−0.058	0.98	−0.21	0.374
Bitcoin	2022–2026	1 722	−0.026	1.01	0.18	0.924

- BET: strong momentum in 1997–2006 ($\hat{\rho}(1) = 0.270$), much weaker afterwards; since 2017 VR(5) is still above 1, but the joint test does not reject
- S&P 500: the negative autocorrelation appears after 2002; Bitcoin: no rejection in any period since 2014; [Urquhart \(2016\)](#) found inefficiency in its earliest years

Why the Early BET Was Predictable

- **Thin trading:** few trades per day in many index stocks
 - ▶ a closing price may be hours old; news enters the index over several days
 - ▶ the index then shows positive autocorrelation even if each stock is efficient
- Changes after 2007
 - ▶ more liquidity, foreign investors, EU membership, the FTSE Russell upgrade to Secondary Emerging (2020) [FTSE Russell \(2019\)](#)
- Crashes on the Bucharest Stock Exchange: [Pele & Mazurencu-Marinescu \(2012\)](#)



The Stock Exchange Palace, Bucharest

Photo: [Neoclassicism Enthusiast \(2019\)](#); CC BY-SA
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Recap: Efficiency across Markets and over Time

- Over the last ten years most markets, including Bitcoin, are consistent with a random walk
- Efficiency changes over time: the AMH; rolling windows show it, but overlap
- The early BET: thin trading and momentum; today: much closer to a random walk
- Many markets, many tests: expect some false rejections

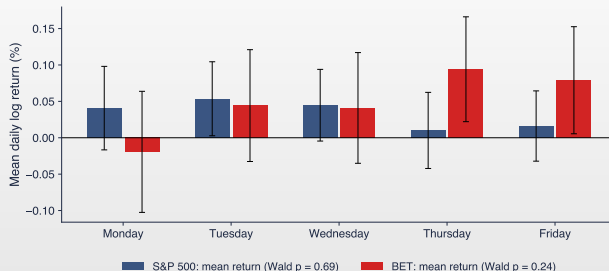
Event Studies and the Strong Form

- **Event study** (semi-strong form) [MacKinlay \(1997\)](#), [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 4: how fast does a price absorb public news?
 - ▶ abnormal return: the return minus the return of a model (for example the market); cumulated over a window around the announcement
 - ▶ efficient: a jump on the day of the news, no drift afterwards
 - ▶ recent research: how fast futures prices absorb news, measured in nanoseconds at Eurex and CME [Tak, Schlamp, Osterrieder & Pele \(2026\)](#); market responses to Ethereum upgrades [Lin, Găman, Wang & Pele \(2025\)](#)
- **Strong form**: can anyone with private information earn excess returns?
 - ▶ insiders (managers trading their own shares) do: insider trading is illegal for this reason
 - ▶ professional fund managers mostly do not [Jensen \(1968\)](#), [Fama & French \(2010\)](#)

Calendar Anomalies

- An **anomaly**: a pattern in returns that the equilibrium model does not explain
 - ▶ **weekend effect**: low or negative Monday returns French (1980)
 - ▶ **January effect**: higher returns in January, mostly for small stocks Rozeff & Kinney (1976)
 - ▶ **turn-of-the-month effect**: returns concentrated around the first trading days of the month Ariel (1987)
- Do anomalies last?
 - ▶ returns of published anomalies fall after publication: traders learn McLean & Pontiff (2016)
 - ▶ hundreds of patterns have been tested: a new one needs $|t| > 3$, not 2 Harvey, Liu & Zhu (2016)
- Data mining: with enough calendar splits, some will look significant by chance

Day-of-Week Returns: S&P 500 and BET



- Mean daily return by weekday with 95% intervals (Newey–West standard errors [Newey & West \(1987\)](#)); Wald test of equal means: S&P 500 $p = 0.69$, BET $p = 0.24$
- BET Monday: -0.019% , the only negative mean, yet not significant; January: BET 0.185% per day against 0.036% ($t = 1.72$, $p = 0.09$); S&P 500: $p = 0.59$
- No calendar effect survives at 5% in our data

Recap: Semi-Strong Form, Strong Form and Anomalies

- Event studies: prices react to public news at once, with no drift, in efficient markets
- Strong form fails for insiders, holds roughly for professional managers
- Calendar anomalies: weak in our data, and they fade after publication

An Open Question

- **Did the BET become more efficient after Romania's upgrade to Secondary Emerging market (September 2020)?** [FTSE Russell \(2019\)](#)
 - ▶ five years before: $\hat{\rho}(1) = -0.002$, $VR(5) = 1.12$ ($Z^* = 0.80$); five years after: $\hat{\rho}(1) = 0.087$, $VR(5) = 1.18$ ($Z^* = 1.78$)
 - ▶ difference of the two $VR(5)$: $z = 0.34$: no evidence of a change
- Why it is open: COVID-19, new large listings and higher volumes happened at the same time; is the index or each stock the right unit?
- AI tools can speed up such a study; they do not replace checking it [Wang et al. \(2023\)](#)

 [SFM_ch7_adaptive_markets](#)

How AI Could Help

- ▣ **Literature:** list studies of efficiency in Central and Eastern European markets and the tests they use
- ▣ **Code:** a first draft of rolling Lo-MacKinlay and automatic VR tests for BET stocks, with wild-bootstrap p-values
- ▣ **Robustness:** other windows, other horizons, BET-TR instead of BET, trading volume as a control
- ▣ Example prompt
 - ▶ Write a Python function that computes the Lo-MacKinlay variance ratio $VR(5)$ and the heteroskedasticity-robust z-statistic for daily BET log returns on rolling 500-day windows, and test whether the mean VR differs before and after 21 September 2020.

What to Check

- ▣ The statistic: robust Z^* , not the homoskedastic Z ; a draft that uses Z will “find” inefficiency
- ▣ Log returns of the index, not price levels; no unit-root test on returns used as an efficiency test
- ▣ Rolling windows overlap: a t -test on the mean of rolling statistics treats dependent numbers as independent
- ▣ The event date and the window are fixed before looking at the result
- ▣ References: every cited paper must exist; check the DOI

Project Idea

- **Question:** is the Bucharest market more efficient today than in 2015?
 - ▶ a second question: does the change appear in the index or in the individual stocks?
 - ▶ data: BET since 1997, BET-TR since 2014, BVB stocks (TLV, SNP, BRD, TGN) since 2010 (EODHD)
- **Steps**
 - ▶ rolling VR(2), VR(5) with Z^* and the Chow–Denning test for the index and for each stock
 - ▶ compare 2015–2020 with 2020–2025 by a block bootstrap of the difference
 - ▶ repeat for the WIG20 and the BUX as controls: did all of them change at the same time?
- **Deliverable:** one table, one chart, and a paragraph on what the data can and cannot show
- **Declare any AI use, and list the errors of the AI that you corrected**

Key Takeaways

- ▣ Efficient market: prices reflect information; weak form = past prices cannot be used to earn excess returns
- ▣ The right null is the martingale (RW3), not RW1: volatility clustering is not inefficiency
- ▣ Log prices are $I(1)$ and returns $I(0)$ in all our series; a unit root alone does not prove efficiency
- ▣ Use robust tests: $\tilde{Q}$ and $Z^*(q)$; join horizons with Chow–Denning or let the data choose (automatic VR)
- ▣ Evidence: S&P 500 slightly mean-reverting, early BET trending, most markets close to efficient today; efficiency varies over time
- ▣ Statistical predictability is not profit: costs and risk decide

Key Formulas

Quantity	Formula
Random walk	$p_t = \mu + p_{t-1} + \varepsilon_t, \quad \text{Var}(r_t(q)) = q\sigma^2$
Sample ACF	$\hat{\rho}(k) = \sum_t e_t e_{t-k} / \sum_t e_t^2, \quad \text{band } \pm 1.96 \sqrt{\hat{\delta}(k)/T}$
Ljung–Box	$Q_{LB}(m) = T(T+2) \sum_{k=1}^m \hat{\rho}(k)^2 / (T-k) \sim \chi^2(m)$
Runs test	$E[R] = 2n_1 n_2 / n + 1, \quad z = (R - E[R]) / \text{sd}(R)$
ADF	$\Delta p_t = c + bt + \gamma p_{t-1} + \sum_j \delta_j \Delta p_{t-j} + \varepsilon_t, \quad \tau = \hat{\gamma} / \text{SE}(\hat{\gamma})$
KPSS	$\sum_t S_t^2 / (T^2 \hat{\lambda}^2), \quad H_0: \text{stationarity}$
VR	$\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho(k)$
Robust VR test	$Z^*(q) = (\widehat{\text{VR}}(q) - 1) / \sqrt{\hat{\theta}(q)/T}, \quad \hat{\theta}(q) = \sum_{j < q} [2(q-j)/q]^2 \hat{\delta}(j)$
Chow–Denning	$\max_i Z^*(q_i) , \quad P(\text{CD} \leq c) = (2\Phi(c) - 1)^m$

Check Yourself

- **Question:** $\hat{\rho}(1) = 0.10$ and $\hat{\rho}(k) = 0$ for $k \geq 2$; what does $VR(2)$ tell us?
 - ▶ **Answer:** $1 + 0.10 = 1.10$: the two-day variance is 10% larger than under a random walk: momentum
- **Question:** ADF does not reject on log prices and KPSS rejects; is the market efficient?
 - ▶ **Answer:** we only know that prices are $I(1)$; efficiency needs tests on the increments (ACF, VR)
- **Question:** $Q_{LB}(10)$ rejects strongly for squared returns; is weak-form efficiency rejected?
 - ▶ **Answer:** no; this is volatility clustering; it rejects RW1 and RW2, not the martingale
- Next: Chapter 8, volatility estimators and volatility clustering

References (1/3)

- ▣ Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- ▣ Ariel, R.A. (1987). A monthly effect in stock returns. *Journal of Financial Economics*, 18(1), 161–174.
- ▣ Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École normale supérieure*, 17, 21–86.
- ▣ Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- ▣ Box, G.E.P., Pierce, D.A. (1970). Distribution of residual autocorrelations in autoregressive-integrated moving average time series models. *Journal of the American Statistical Association*, 65(332), 1509–1526.
- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- ▣ Choi, I. (1999). Testing the random walk hypothesis for real exchange rates. *Journal of Applied Econometrics*, 14(3), 293–308.
- ▣ Chow, K.V., Denning, K.C. (1993). A simple multiple variance ratio test. *Journal of Econometrics*, 58(3), 385–401.
- ▣ Dickey, D.A., Fuller, W.A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431.
- ▣ Escanciano, J.C., Lobato, I.N. (2009). An automatic Portmanteau test for serial correlation. *Journal of Econometrics*, 151(2), 140–149.
- ▣ Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.
- ▣ Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- ▣ Fama, E.F. (1991). Efficient capital markets: II. *Journal of Finance*, 46(5), 1575–1617.
- ▣ Fama, E.F., French, K.R. (2010). Luck versus skill in the cross-section of mutual fund returns. *Journal of Finance*, 65(5), 1915–1947.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ French, K.R. (1980). Stock returns and the weekend effect. *Journal of Financial Economics*, 8(1), 55–69.

References (2/3)

- ▣ FTSE Russell (2019). [FTSE Russell announces results of country classification review](#). Press release, London Stock Exchange Group.
- ▣ Granger, C.W.J., Newbold, P. (1974). [Spurious regressions in econometrics](#). *Journal of Econometrics*, 2(2), 111–120.
- ▣ Grossman, S.J., Stiglitz, J.E. (1980). [On the impossibility of informationally efficient markets](#). *American Economic Review*, 70(3), 393–408.
- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). [... and the cross-section of expected returns](#). *Review of Financial Studies*, 29(1), 5–68.
- ▣ Jensen, M.C. (1968). [The performance of mutual funds in the period 1945–1964](#). *Journal of Finance*, 23(2), 389–416.
- ▣ Kendall, M.G. (1953). [The analysis of economic time-series, Part I: prices](#). *Journal of the Royal Statistical Society, Series A*, 116(1), 11–34.
- ▣ Kim, J.H. (2009). [Automatic variance ratio test under conditional heteroskedasticity](#). *Finance Research Letters*, 6(3), 179–185.
- ▣ Kim, J.H., Shamsuddin, A., Lim, K.-P. (2011). [Stock return predictability and the adaptive markets hypothesis: evidence from century-long U.S. data](#). *Journal of Empirical Finance*, 18(5), 868–879.
- ▣ Kwiatkowski, D., Phillips, P.C.B., Schmidt, P., Shin, Y. (1992). [Testing the null hypothesis of stationarity against the alternative of a unit root](#). *Journal of Econometrics*, 54(1–3), 159–178.
- ▣ Lim, K.-P., Brooks, R. (2011). [The evolution of stock market efficiency over time: a survey of the empirical literature](#). *Journal of Economic Surveys*, 25(1), 69–108.
- ▣ Lin, Y., Găman, A., Wang, Y., Pele, D.T. (2025). [Market responses to Ethereum development milestones: an event study approach](#). SSRN Working Paper 5176758.
- ▣ Ljung, G.M., Box, G.E.P. (1978). [On a measure of lack of fit in time series models](#). *Biometrika*, 65(2), 297–303.
- ▣ Lo, A.W. (2004). [The adaptive markets hypothesis](#). *Journal of Portfolio Management*, 30(5), 15–29.
- ▣ Lo, A.W. (2017). [Adaptive Markets: Financial Evolution at the Speed of Thought](#). Princeton University Press.
- ▣ Lo, A.W., MacKinlay, A.C. (1988). [Stock market prices do not follow random walks: evidence from a simple specification test](#). *Review of Financial Studies*, 1(1), 41–66.
- ▣ MacKinlay, A.C. (1997). [Event studies in economics and finance](#). *Journal of Economic Literature*, 35(1), 13–39.

References (3/3)

- ▣ MacKinnon, J.G. (1996). Numerical distribution functions for unit root and cointegration tests. *Journal of Applied Econometrics*, 11(6), 601–618.
- ▣ Malkiel, B.G. (2003). The efficient market hypothesis and its critics. *Journal of Economic Perspectives*, 17(1), 59–82.
- ▣ McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Pele, D.T., Mazurencu-Marinescu, M. (2012). Modelling stock market crashes: the case of Bucharest Stock Exchange. *Procedia – Social and Behavioral Sciences*, 58, 533–542.
- ▣ Phillips, P.C.B., Perron, P. (1988). Testing for a unit root in time series regression. *Biometrika*, 75(2), 335–346.
- ▣ Roberts, H.V. (1959). Stock-market “patterns” and financial analysis: methodological suggestions. *Journal of Finance*, 14(1), 1–10.
- ▣ Rozeff, M.S., Kinney, W.R. (1976). Capital market seasonality: the case of stock returns. *Journal of Financial Economics*, 3(4), 379–402.
- ▣ Said, S.E., Dickey, D.A. (1984). Testing for unit roots in autoregressive-moving average models of unknown order. *Biometrika*, 71(3), 599–607.
- ▣ Samuelson, P.A. (1965). Proof that properly anticipated prices fluctuate randomly. *Industrial Management Review*, 6(2), 41–49; reprinted in the *World Scientific Handbook in Financial Economics Series*, 25–38.
- ▣ Tak, A., Schlamp, S., Osterrieder, J., Pele, D.T. (2026). Racing the news: nanosecond market efficiency at Eurex and CME. SSRN Working Paper 7157232.
- ▣ Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley.
- ▣ Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80–82.
- ▣ Urquhart, A., McGroarty, F. (2016). Are stock markets really efficient? Evidence of the adaptive market hypothesis. *International Review of Financial Analysis*, 47, 39–49.
- ▣ Wald, A., Wolfowitz, J. (1940). On a test whether two samples are from the same population. *Annals of Mathematical Statistics*, 11(2), 147–162.
- ▣ Wang, H., Fu, T., Du, Y., Gao, W., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620, 47–60.